AI-Powered Wildfire Early Detection & Alerting System

MILESTONE 2 — DATASET PREPARATION

Multi-Source Remote Sensing Dataset Preparation for Deep Learning

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Project Objective

Develop a **multi-source geospatial dataset** for wildfire prediction by integrating satellite imagery, weather, atmospheric, and terrain data into a unified daily fused product for deep learning.

Data Sources

- **Sentinel-2** — Vegetation health indices
- **Landsat-8** — Thermal and optical features
- **ERA5** — Weather and climate variables
- **Sentinel-5P** — Atmospheric conditions
- **Copernicus DEM** — Terrain features

Ground Truth

NASA FIRMS Active Fire detections

Output

Daily fused dataset ready for deep learning model training

Multi-Source Dataset Overview

NASA FIRMS

Binary fire / no-fire labels

Sentinel-2

Vegetation health indices

Landsat-8

Thermal + optical features

ERA5

14 weather variables

Sentinel-5P

Atmospheric conditions

Copernicus DEM

6 terrain layers

Area of Interest: California · Time Period: 2016–2025 · Prediction Target: Binary Fire (>=30% Confidence score) / No-Fire (<30%)

Dataset Summary

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Data Sources

Integrated remote sensing and climate datasets

3642

Daily FIRMS Labels

Active fire detections across the study period

480

Sentinel-2 Tiles

High-resolution optical imagery

264

Landsat-8 Tiles

Thermal and multispectral coverage

91

Sentinel-5P Composites

Monthly atmospheric data products

14

ERA5 Variables

Weather and climate parameters

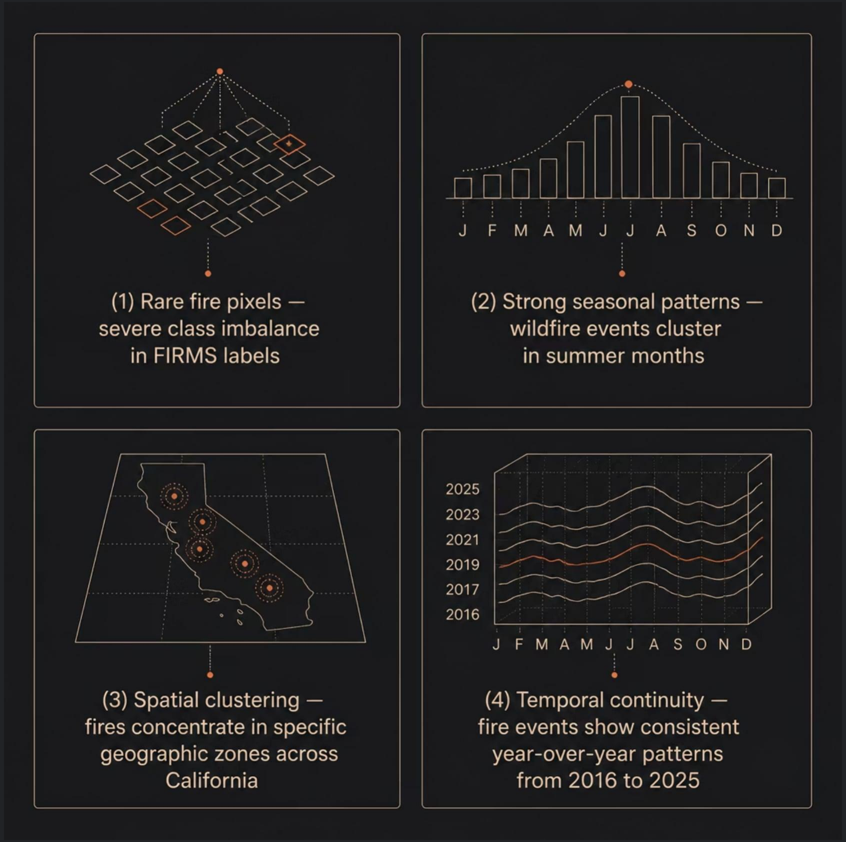
32

Feature Channels

Final fused tensor depth for model input

Coverage spans California from 2016–2025, with spatial resolutions ranging from **10 m** (Sentinel-2) to **1 km** (ERA5), unified onto a common reference grid.

Exploratory Data Analysis — Key Findings



FIRMS Analysis

- Fire pixels are rare — severe class imbalance confirmed
- Fire events exhibit strong temporal seasonality
- Spatial clustering observed across California regions

Dataset Insights

- Peak wildfire activity concentrated in summer months
- Temporal continuity validated across 2016–2025
- Class imbalance motivates weighted and focal loss functions during model training

Source-Wise Feature Findings

NASA FIRMS
Confidence filtering applied; binary fire / no-fire label generation

Landsat-8
NDVI and Land Surface Temperature (LST) derived from thermal and optical bands

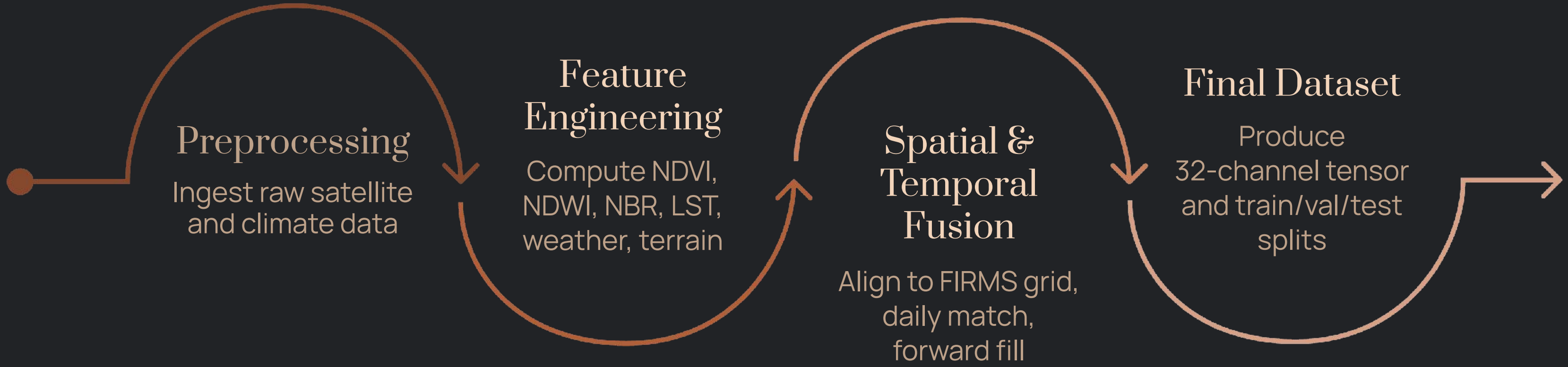
Sentinel-2
NDVI, NDWI, and NBR indices for vegetation health and burn severity

ERA5
Temperature, wind, humidity, rainfall, and soil moisture — 14 variables

Sentinel-5P
Aerosol Index capturing atmospheric smoke and particulate conditions

Copernicus DEM
Elevation, slope, aspect, hillshade, TRI, and TPI — 6 terrain layers

Feature Extraction & Data Fusion Pipeline



The fusion pipeline aligns all six data sources to a common **FIRMS reference grid** (spatial) and applies daily matching with forward filling and monthly broadcasting (temporal), producing a unified 32-channel tensor ready for model input.

Train / Validation / Test Split Strategy

Temporal Split Configuration

Train — 70% 2018–2022	**Validation — 15%** 2023–2024
Test — 15% 2025	

Split Design Principles

- **Temporal split** — preserves chronological integrity; no future data leakage
- **7-day sliding window** — captures temporal context for each prediction sample
- **Strict separation** — validation and test periods are held out entirely from training

Temporal splitting ensures the model is evaluated on genuinely unseen future conditions, simulating real-world deployment.

Challenges & Solutions

Challenge	Solution
Landsat coverage gaps	Forward fill strategy applied
Missing cloud mask	Documented as known limitation
Monthly Sentinel-5P frequency	Monthly broadcast to daily grid
Severe class imbalance	Weighted loss and focal loss
Multi-resolution datasets	Common FIRMS reference grid
Multi-temporal alignment	Daily temporal alignment pipeline

Final Dataset & Next Milestone

1

Milestone 2 — Complete

32-channel dataset: X_train.npy, y_train.npy, X_val.npy, y_val.npy, X_test.npy, y_test.npy

2

Milestone 3 — Model Development

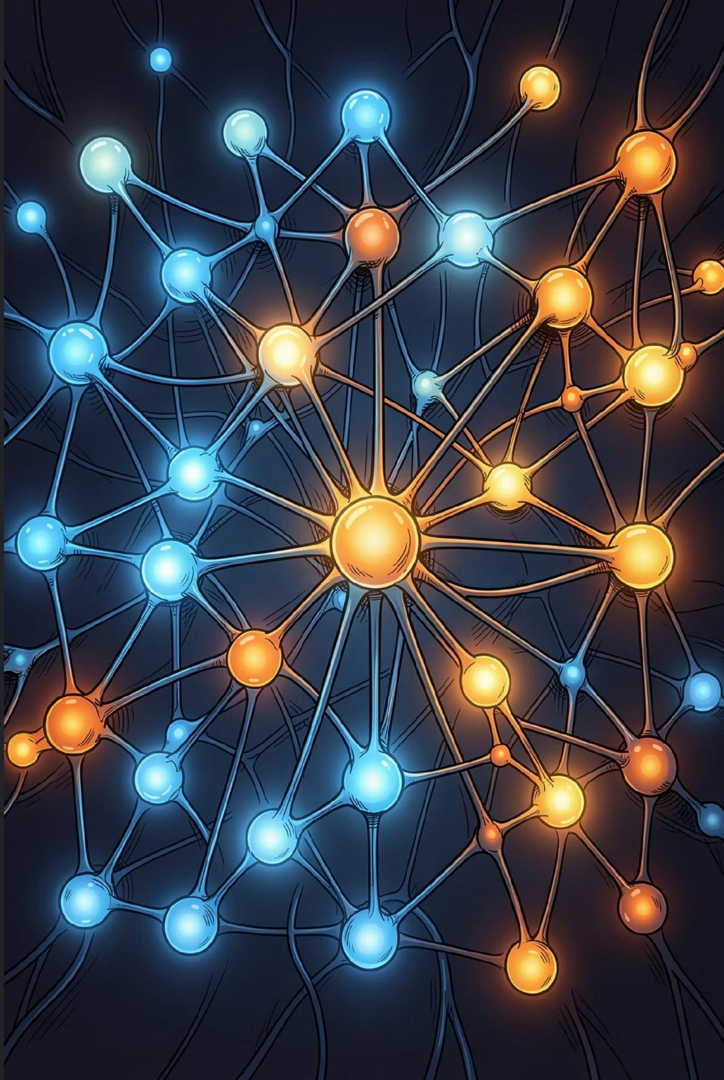
CNN / Swin Transformer with temporal feature learning and multi-source fusion

3

Milestone 4 — Early Warning

Wildfire risk prediction scores and early warning alert generation

Milestone 2 dataset is ready. The foundation is set for multimodal deep learning model development in Milestone 3.



Signatures

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